

1. Write Nested Loops for Grids and Pattern Problems

What is a Nested Loop?

A **nested loop** is a loop inside another loop.

```
for(int i = 0; i < 3; i++) {  
    for(int j = 0; j < 4; j++) {  
        System.out.print("*");  
    }  
    System.out.println();  
}
```

Output

```
****  
****  
****
```

Outer loop → Rows

Inner loop → Columns

Think of a matrix.

```
(0,0) (0,1) (0,2)  
(1,0) (1,1) (1,2)  
(2,0) (2,1) (2,2)
```

The outer loop changes the row.

The inner loop changes the column.

Golden Rule

Outer loop → Number of Rows

Inner loop → Number of Columns

Never confuse these.

Example 1

Print

```
123  
123  
123
```

```
for(int i = 1; i <= 3; i++) {  
    for(int j = 1; j <= 3; j++) {  
        System.out.print(j);  
    }  
    System.out.println();  
}
```

Example 2

```
*  
**  
***  
****
```

```
for(int i = 1; i <= 4; i++) {  
    for(int j = 1; j <= i; j++) {  
        System.out.print("*");  
    }  
}
```

```
    System.out.println();  
}
```

Notice

```
Row 1 → 1 star  
  
Row 2 → 2 stars  
  
Row 3 → 3 stars
```

The inner loop depends on the outer loop.

Example 3

```
****  
***  
**  
*
```

```
for(int i = 4; i >= 1; i--) {  
    for(int j = 1; j <= i; j++) {  
        System.out.print("*");  
    }  
    System.out.println();  
}
```

Matrix Traversal

Very common interview question.

```
int[][] arr = {  
    {1,2,3},
```

```
{4,5,6},  
{7,8,9}  
};  
  
for(int i=0;i
```

Hard Interview Question

Print only the diagonal

```
1 0 0  
0 5 0  
0 0 9
```

```
if(i==j)
```

Secondary diagonal

```
0 0 3  
0 5 0  
7 0 0
```

Condition

```
i + j == n - 1
```

Border Elements

Condition

```
if(i==0 || j==0 || i==n-1 || j==m-1)
```

Very common.

Time Complexity

```
for(...)  
  for(...)
```

Rows × Columns

$O(n \times m)$

If square matrix

$O(n^2)$

Interview Trap

Q:

```
for(i=0;i
```

Complexity?

Answer

```
n+(n-1)+(n-2)...  
  
= n(n+1)/2  
  
=  $O(n^2)$ 
```

NOT $O(n)$.

2. Trace Loop Variables Manually Using Dry Run Tables

Interviewers LOVE this.

You should be able to predict output without running code.

Example

```
for(int i=1;i<=3;i++){  
    System.out.print(i+" ");  
}
```

Dry Run

Iteration	i	Condition	Output
1	1	True	1
2	2	True	2
3	3	True	3
4	4	False	Stop

Output

```
1 2 3
```

More Difficult

```
int sum=0;  
for(int i=1;i<=4;i++){
```

```
    sum+=i;  
}
```

Dry Run

i	sum
1	1
2	3
3	6
4	10

Nested Loop Dry Run

```
for(int i=1;i<=2;i++){  
    for(int j=1;j<=3;j++){  
        System.out.print(i+" "+j+" ");  
    }  
}
```

Table

```
i=1  
j=1 →11  
j=2 →12  
j=3 →13  
i=2  
j=1 →21  
j=2 →22
```

```
j=3 →23
```

Output

```
11 12 13 21 22 23
```

Hard Interview

Predict

```
for(int i=0;i<3;i++){  
    i++;  
    System.out.print(i+" ");  
}
```

Dry Run

Iteration 1

```
i=0  
inside loop  
i++  
becomes 1  
print 1  
for update  
i++  
becomes 2
```

Second iteration

```
i=2  
inside  
i++  
3  
print 3  
update  
4  
stop
```

Output

```
1 3
```

Most freshers answer

```
1 2 3
```

Wrong.

3. Understand Off-by-One Errors in Loop Boundaries

One of the most common programming bugs.

Example

```
for(int i=0;i<=5;i++)
```

Runs

```
0 1 2 3 4 5
```

Total

```
6 times
```

Many people expect 5.

Correct

```
for(int i=0;i<5;i++)
```

Runs

```
0 1 2 3 4
```

Exactly 5 times.

Array Example

```
int arr[]={10,20,30};
```

Correct

```
for(int i=0;i
```

Wrong

```
for(int i=0;i<=arr.length;i++)
```

When

```
i=3
```

It becomes

```
arr[3]
```

Exception

```
ArrayIndexOutOfBoundsException
```

Binary Search Bug

Almost every beginner writes

```
while(low
```

instead of

```
while(low<=high)
```

Missing

The last element.

String Example

```
String s="JAVA";
```

Correct

```
for(int i=0;i
```

Wrong

```
i<=s.length()
```

Exception

```
StringIndexOutOfBoundsException
```

Interview Question

Difference

```
i
```

vs

```
i<=n
```

Answer

4. Calculate Simple Time Complexity for One Loop and Nested Loops

Interviewers ask this constantly.

Single Loop

```
for(int i=0;i
```

Runs

n times

Complexity

$O(n)$

Loop Increment by 2

```
for(int i=0;i
```

Runs

$n/2$

Still

$O(n)$

Ignore constants.

Loop Multiplication

```
for(int i=1;i
```

1

2

4

8

16

How many times?

$\log_2 n$

Complexity

$O(\log n)$

Nested Loop

for(i=0;i

$n \times n$

$O(n^2)$

Rectangular Matrix

for(i=0;i

$O(nm)$

Triangular Loop

```
for(int i=0;i
```

Iterations

```
n+(n-1)+(n-2)...  
=n(n+1)/2
```

Still

```
O(n2)
```

Independent Loops

```
for(...n...)  
for(...n...)
```

Complexity

```
O(n+n)  
  
=  
  
O(n)
```

Not $O(n^2)$, because the loops are **sequential**, not nested.

Hard Interview Questions

Q1

```
for(int i=0;i
```

Answer:

Inner loop runs $0 + 1 + 2 + \dots + (n-1) = n(n-1)/2$

Complexity: $O(n^2)$

Q2

```
for(int i=1;i
```

Answer: $O(\log n)$

Q3

```
for(int i=0;i
```

Answer: Outer loop is $O(n)$, inner loop is $O(\log n)$

Overall: $O(n \log n)$

Q4

```
for(int i=0;i
```

Answer: $O(n^3)$

Q5

```
for(int i=n;i>0;i/=2)
```

Answer: $O(\log n)$

Key Interview Takeaways

- **Nested loops:** Think in terms of **rows and columns**. Practice patterns and matrix traversal.
- **Dry runs:** Trace variables step by step; interviewers often ask you to predict output without executing code.
- **Off-by-one errors:** Be precise with $<$ vs $\leq$, especially for arrays, strings, and binary search.
- **Time complexity:**
 - $i++$ or $i += k \rightarrow O(n)$
 - $i *= 2$ or $i /= 2 \rightarrow O(\log n)$
 - Nested loops $\rightarrow$ Multiply complexities
 - Sequential loops $\rightarrow$ Add complexities, then simplify

Mastering these four topics will make many entry-level Java and DSA interview questions much easier to analyze and solve.